

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460730.7833 +/- 0.0058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.049
Transit depth (Rp/Rs)^2	2.76 +/- 1.63 [%]
Orbital Inclination (inc)	87.77 +/- 2.83
Airmass coefficient 1 (a1)	0.093 +/- 0.055
Airmass coefficient 2 (a2)	1.62 +/- 0.71
Scatter in the residuals of the lightcurve fit is	50.92 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	7.5
Transit Duration (day)	0.0933 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.049 vs 0.1368 (deviation 0.53σ)
Duration fit vs published	0.0933 d vs 0.0758 d (deviation 1.82σ)
Mid-time O–C	13.1 min
Depth SNR	3.0
Residual scatter	50.92 %

Light curve

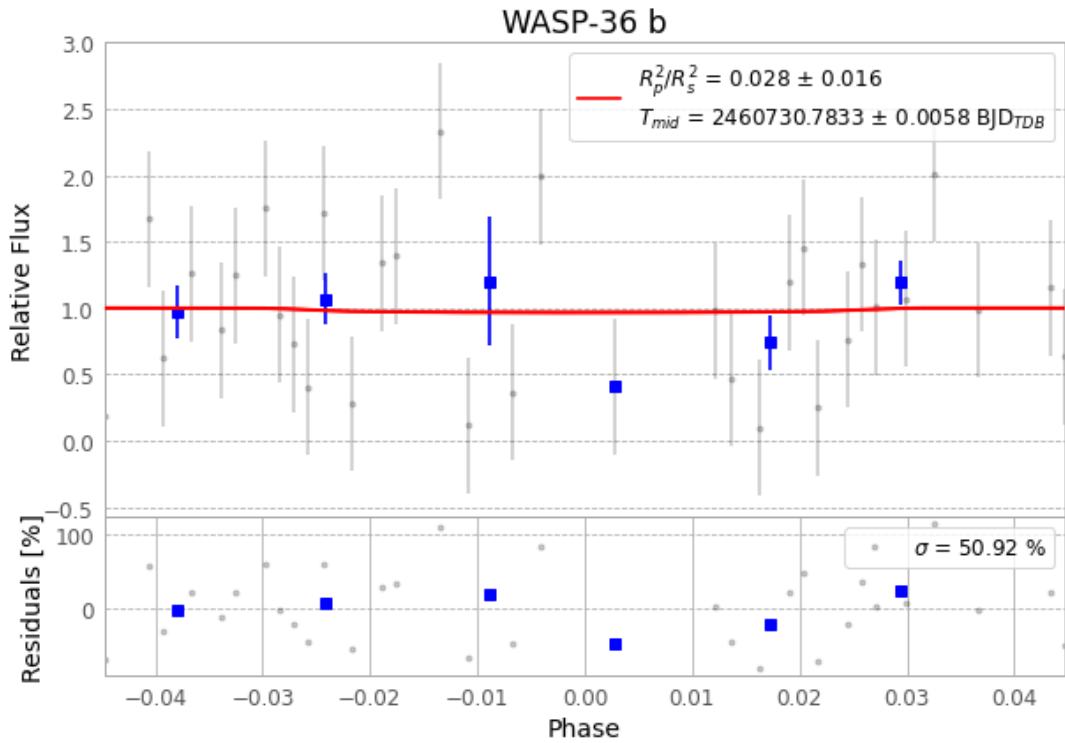


Figure 1 — FinalLightCurve_WASP-36 b_2025-02-23.png (SHA-256 a58584327d58ffa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [524, 396], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 a144f0c8a3156a63...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

69 FITS frames (45 MB), WASP-36250224045415.FITS → WASP-36250224082116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-250224.log (42efc9ce22a3...), FinalLightCurve_WASP-36 b_2025-02-23.png (a58584327d58...), FinalLightCurve_WASP-36 b_2025-02-23.csv (66ead7014c86...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 02:50Z.